CYCLISTIC 2025 REPORT

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SUMMARY

INTRODUCTION

This study aims to understand the behavioral differences between casual users and members of Cyclistic, a bike-sharing service. The objective is to identify opportunities to encourage more casual users to become members through data-driven insights and targeted strategies.

ABOUT THE COMPANY

In 2016, Cyclistic launched a successful bike-share offering. Since then, the program has grown to a fleet of 5,824 bicycles that are geotracked and locked into a network of 692 stations across Chicago. The bikes can be unlocked from one station and returned to any other station in the system anytime.

Until now, Cyclistic's marketing strategy relied on building general awareness and appealing to broad consumer segments. One approach that helped make these things possible was the flexibility of its pricing plans: single-ride passes, full-day passes, and annual memberships. Customers who purchase single-ride or full-day passes are referred to as casual riders. Customers who purchase annual memberships are Cyclistic members.

Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders. Although the pricing flexibility helps Cyclistic attract more customers, Moreno believes that maximizing the number of annual members will be key to future growth. Rather than creating a marketing campaign that targets all-new customers, Moreno believes there is a solid opportunity to convert casual riders into members. She notes that casual riders are already aware of the Cyclistic program and have chosen Cyclistic for their mobility needs.

Moreno has set a clear goal: Design marketing strategies aimed at converting casual riders into annual members. In order to do that, however, the team needs to better understand how annual members and casual riders differ, why casual riders would buy a membership, and how digital media could affect their marketing tactics. Moreno and her team are interested in analyzing the Cyclistic historical bike trip data to identify trends.

<u>DATASET OVERVIEW</u>

The dataset covers 12 months of Cyclistic bike-sharing service usage in 2025 and contains information related to:

- Ride IDs
- Departure and arrival dates and times
- Departure and arrival station names
- Longitudes and latitudes of the departure and arrival stations
- Bike types
- User categories (casual and member)

The dataset was used to explore user behavior patterns, identify differences between casual and member riders, and support recommendations for increasing membership adoption.

CLEANING

A sample of data was collected and cleaned in Google Sheets to familiarize ourselves with the manipulations and automations that will be performed later in Python. This sample corresponds to the data from January.

The cleanup performed with *Google Sheets* is as follows:

- Check for duplicates and delete them : ***=countif(range;column) → if >1 = duplicate → Delete***
- Replace empty start and end stations by “Unknown Station”
- Extract departing times: ***=MID(C2;11;9)***
- Extract arrival times: ***=MID(E2;11;9)***
- Calculate the travel time in minutes for each ID: ***=(ended_time-started_time)*1440***
- Extract days of the week: ***=text(started_day;"dddd")***
- Extract months: ***=text(started_day;"mmmm")***
- Identify weekends and weekdays: ***=IF(OR((R2)="saturday";(R2)="sunday");"Week-end";"Weekday")***
- Identify negative travel time values using conditional formatting: ***Select the column > Format > Conditional Formatting > Is Less Than 0 > Red***
- Count the negative values and delete them if they represent only a small portion of the total data: ***=countif(range;column) → Select the column > Add a filter > Condition: less than 0 > Select all rows > Delete rows.***
- Identify travel times greater than 1440 using conditional formatting and delete them.
- Identify excessively short travel times using conditional formatting (less than 1) → Remove the lines using the filter.

After cleaning this sample data, the original dataset was imported into Google Colab to clean the remaining data using the *Python* programming language. Since this database was available on Kaggle, it was imported directly from that platform.

```
import pandas as pd
import kagglehub
path =
kagglehub.dataset_download("ahmedmashal/cyclistics-bike-share-
2023")

import os
print(os.listdir(path))
```

```
import glob
files = glob.glob(path + "/*.csv")
```

```
#Concatenate data
df = pd.concat(
    [
        pd.read_csv(
            f,
            usecols=[
                "ride_id",
              "rideable_type",
              "started_at",
              "ended_at",
              "start_station_name",
              "end_station_name",
              "start_lat",
              "start_lng",
              "member_casual"
                ]
            )
        for f in files
        ],
    ignore_index=True
    )
df.head()
```

```
#Convert dates
df['started_at'] = pd.to_datetime(df["started_at"])
df["ended_at"]= pd.to_datetime(df["ended_at"])
```

```
#Travel time
df["ride_length"] = (df['ended_at'] -
df['started_at']).dt.total_seconds()/60
```

```
#Delete negative ride lengths
df = df[df['ride_length'] >=0]

#Delete rides < 1 min
df = df[df['ride_length'] >=1]
```

Travel times with a negative duration, or less than 1 minute, were removed to avoid anomalies that could bias the analysis. These data were likely to correspond to system errors, cancellations, or unrepresentative journeys.

```
#Fill missing stations
df['start_station_name'] =
df['start_station_name'].fillna("Unknown_Station")
df['end_station_name'] =
df['end_station_name'].fillna("Unknown_Station")
```

```
#Delete duplicate data
df = df.drop_duplicates()
```

```
#Day of week
df['day_of_week'] = df['started_at'].dt.day_name()

#Hour
df['hour'] = df['started_at'].dt.hour

#Month
df['month'] = df['started_at'].dt.month_name()
```

```
#Weekend
df['is_weekend'] = df['started_at'].dt.weekday >= 5
```

EXPLANATORY ANALYSIS

A. User distribution

Question: *“What proportion of users are casual versus members?”*

```
#Member vs Casual distribution
df['member_casual'].value_counts()
```

	count
member_casual	
member	3485110
casual	1920483

```
#Percentage of member and casual
df['member_casual'].value_counts(normalize=True) * 100
```

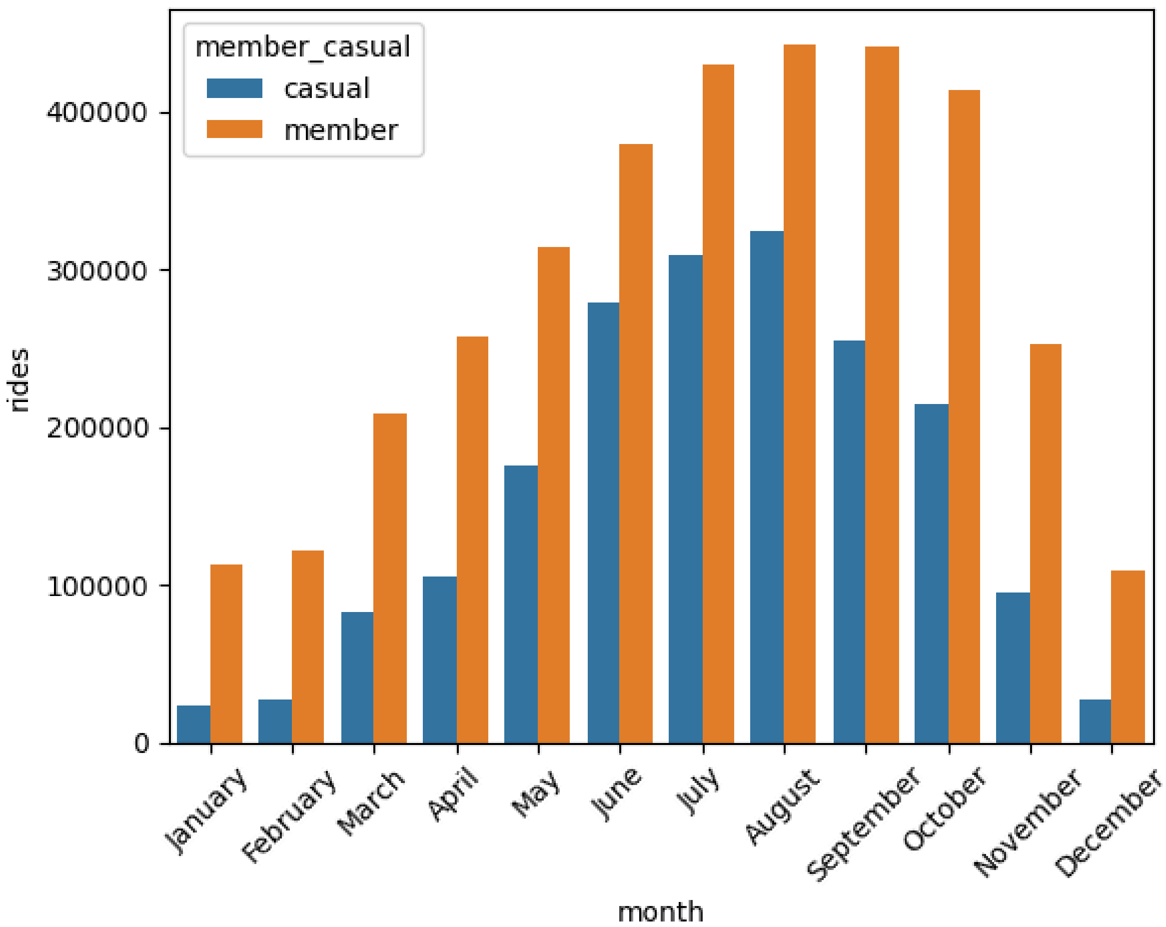
	proportion
member_casual	
member	64.472297
casual	35.527703

➪ **Members represent the majority of total users, suggesting that subscription-based users rely more consistently on the bike for their daily mobility needs. Casual riders account for a smaller share of rides, but still represent a valuable target group for conversion strategies, as they are already engaged in the service.**

B. Seasonality

```
#Monthly rides
month_ride = df.groupby(['month',
'member_casual']).size().reset_index(name='rides')
```

```
#Visualization
month_order = ("January", "February", "March", "April", "May", "June",
"July", "August", "September", "October", "November", "December")
sns.barplot(data = month_ride, x='month', y= 'rides', hue=
'member_casual', order= month_order)
plt.xticks(rotation= 45)
plt.show()
```



➪ **Both user groups show strong seasonality, with usage peaking during summer months and decreasing significantly during winter. Casual riders exhibit a sharper increase between June and August, suggesting stronger recreational and seasonal patterns. Member usage remains comparatively more stable throughout the year, indicating more routine and utilitarian travel behavior.**

C. Travel time

Question: "*Who travels the longest distances?*"

```
#Average cycling trip duration by user type
df.groupby('member_casual')['ride_length'].mean()
```

	ride_length
member_casual	
casual	23.509521
member	12.567106

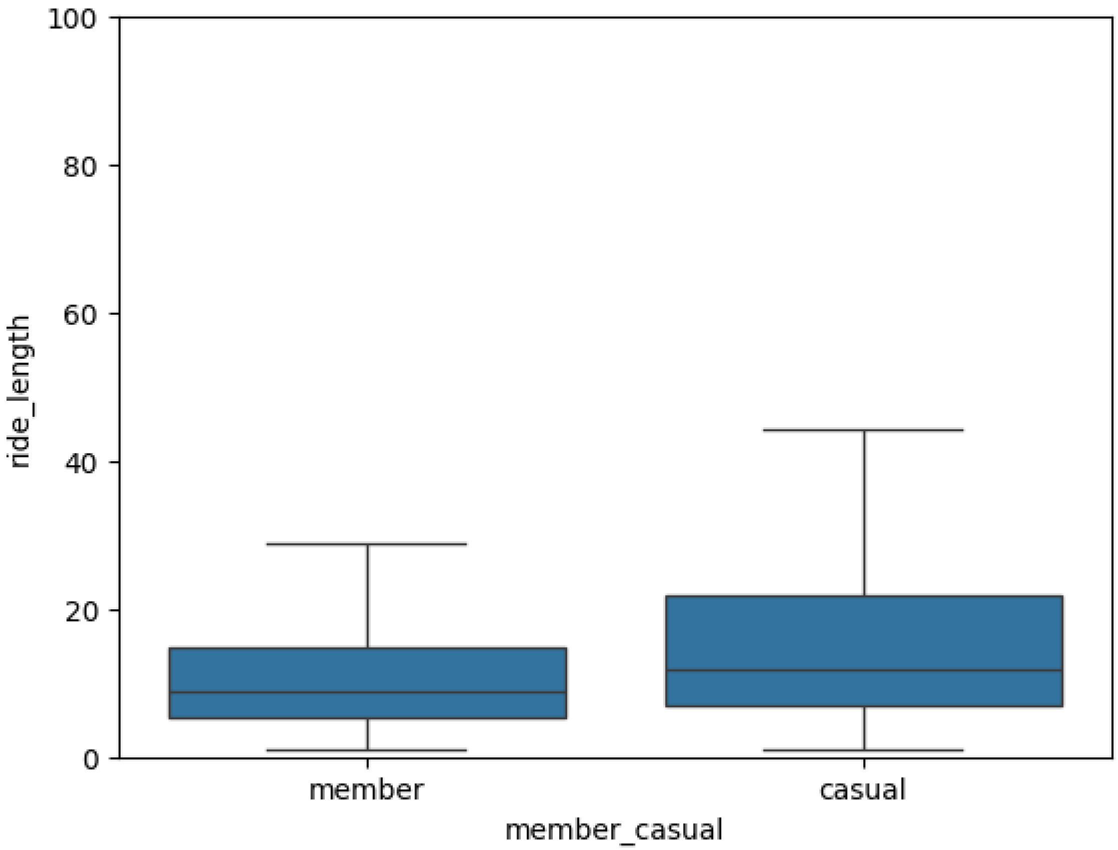
The median duration was also calculated because the average can be skewed by extreme journeys.

```
#Median duration
df.groupby('member_casual')['ride_length'].median()
```

	ride_length
member_casual	
casual	11.924033
member	8.740000

```
#Visualization
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(data = df, x = 'member_casual', y = 'ride_length',
showfliers = False)
plt.ylim(0, 100)
plt.show()
```



➪ **Casual riders exhibit longer average ride duration compared to members, suggesting a more recreational usage pattern. In contrast, members appear to use the service for shorter and more utilitarian trips, likely related to commuting or daily transportation needs.**

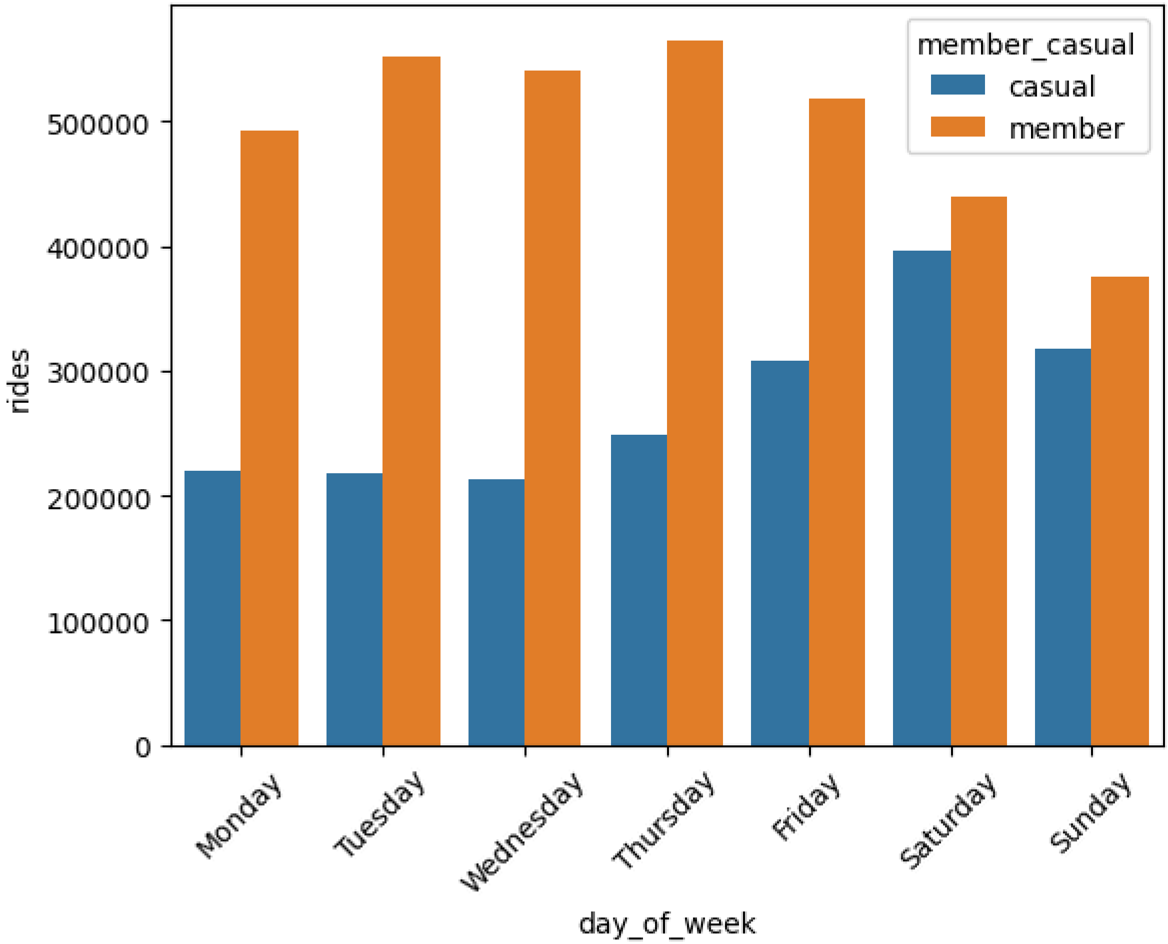
D. Usage per day of the week

Questions: *“What is the proportion of daily use?”*

```
#Number of ride per day
daily_rides = df.groupby(['member_casual',
'day_of_week']).size().reset_index(name='rides')
```

```
#Visualization of usage per day of the week
day_order = ["Monday", "Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday", "Sunday"]
sns.barplot(data = daily_rides, x = 'day_of_week', y =
'rides', hue = 'member_casual', order = day_order)
plt.xticks(rotation=45)
```

```
plt.show()
```



➪ **During the week, members use the service significantly more than casual riders, reinforcing the hypothesis that members primarily use bike-sharing for commuting and daily transportation. The gap between members and casual riders narrows considerably on Saturdays and Sundays, further supporting the idea that casual riders mainly use the service for recreational and leisure activities.**

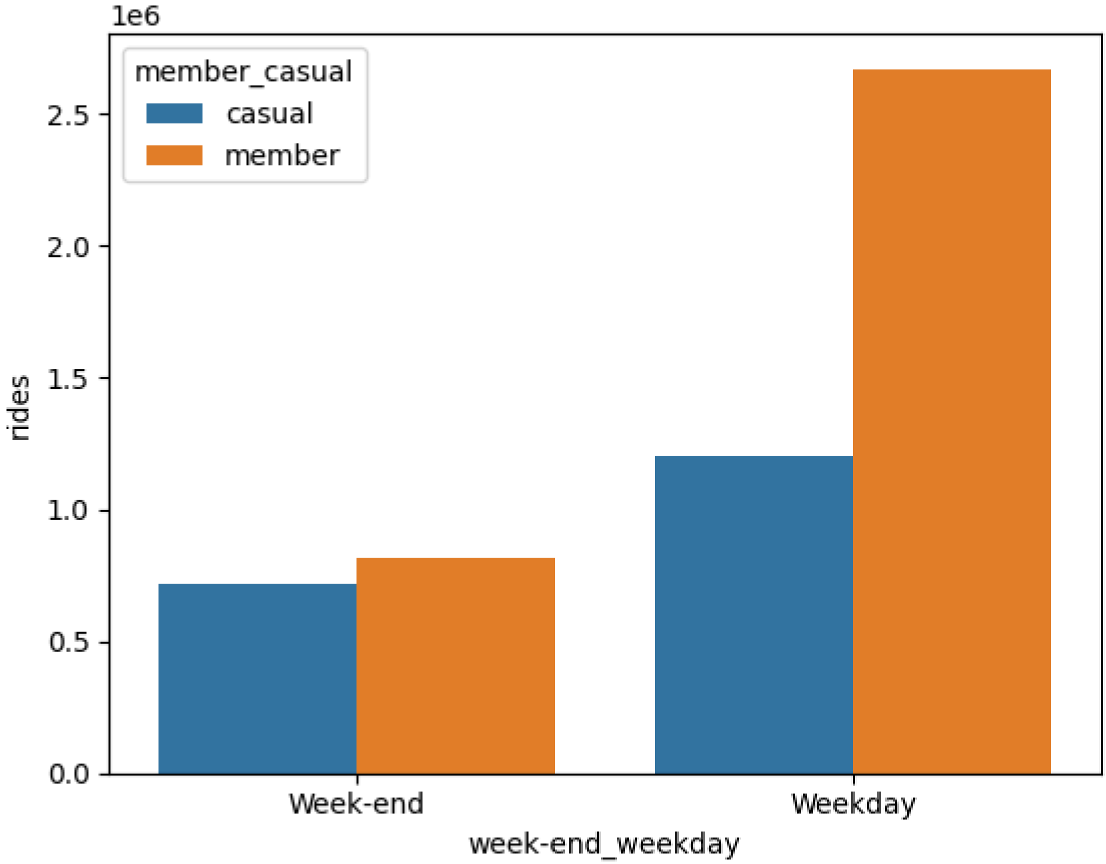
E. <u>Weekend and Weekday comparison</u>

Questions:
- *"Who rides on weekdays?"*
- *"Who rides on weekends?"*

```
#Weekend vs weekday comparison
df['week-end_weekday'] = df['is_weekend'].map({True: "Week-end", False:
"Weekday"})
```

```
week_usage = df.groupby(['week-end_weekday',
'member_casual']).size().reset_index(name='rides')
```

```
#Visualization
sns.barplot(data = week_usage, x="week-end_weekday", y="rides",
hue="member_casual")
plt.show()
```

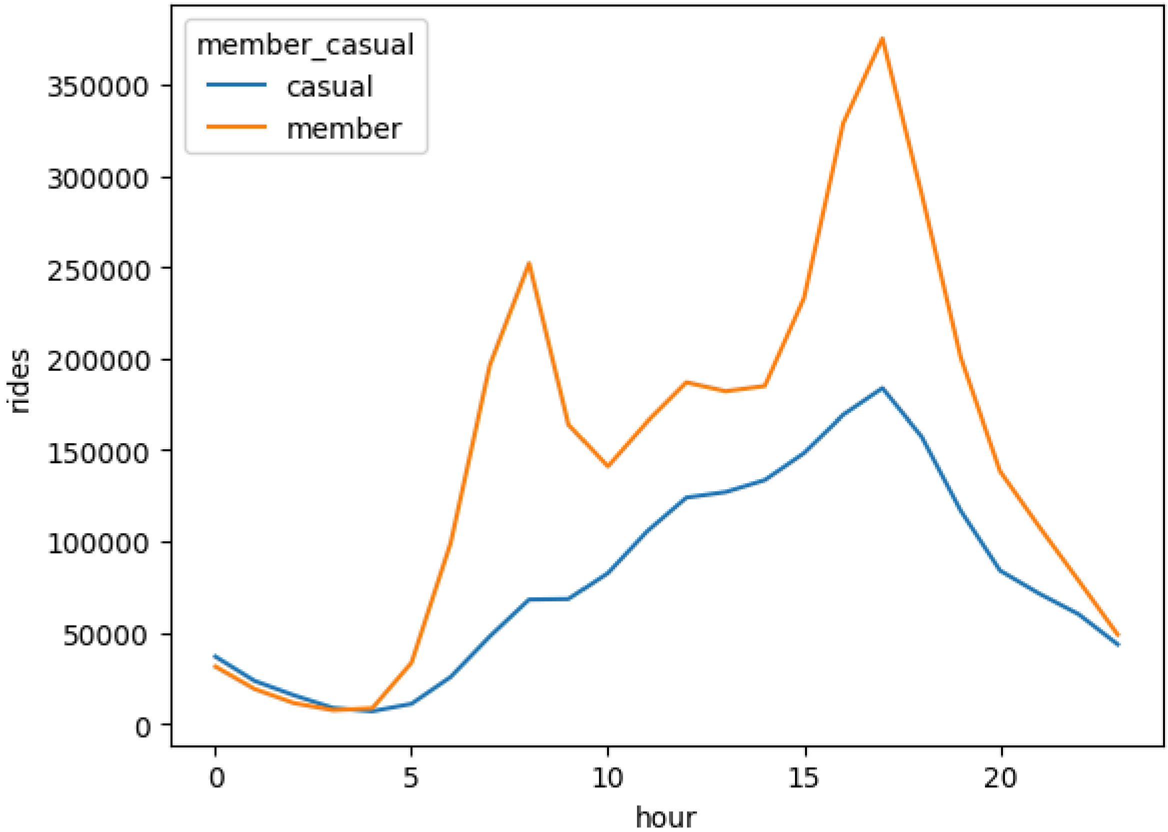


➩ **Members use the service more frequently during weekdays. This trend decreases significantly during weekends, supporting the hypothesis that they primarily use the service for commuting and daily transportation.**
The gap between members and casual users narrows considerably on weekends, suggesting that casual riders are more likely to use the bike-sharing service for leisure and recreational activities.

F. Usage per hour

Question: *"When are bikes used the most throughout the day?"*

```
#Usage per hour
rides_by_hour = df.groupby(['hour',
'member_casual']).size().reset_index(name='rides')
#Visualization
sns.lineplot(data = rides_by_hour, x='hour', y='rides',
hue='member_casual')
plt.show()
```



➩ **Usage peaks among members are observed around 8 a.m. and 5 p.m., corresponding to typical commuting hours. This pattern suggests that members primarily use the service for commuting and daily transportation purposes.**

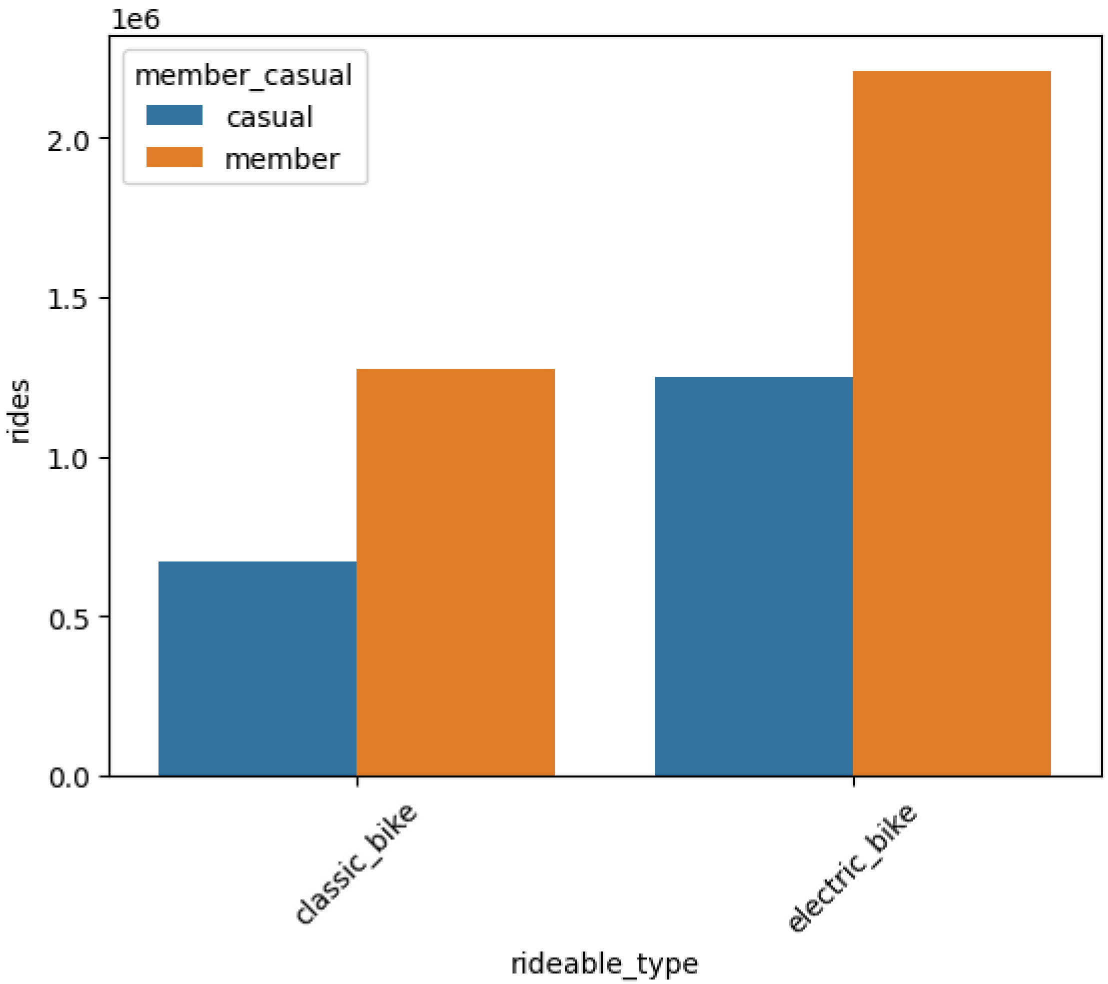
Casual users, on the other hand, show a more evenly distributed usage pattern throughout the day, which may indicate a more recreational and leisure-oriented use of the service.

G. <u>Bike types</u>

Question: *"Which bikes do users prefer?"*

```
#Distribution by type of bikes
bike_type = df.groupby(['rideable_type',
'member_casual']).size().reset_index(name='rides')
```

```
#Visualization
sns.barplot(data = bike_type, x='rideable_type', y='rides',
hue='member_casual')
plt.xticks(rotation=45)
plt.show()
```



➩ **Both member and casual users have a preference for electric bikes.**
It can be suggested that electric bikes are easier to use, faster than classic bikes and less tiring, which is very practical on a daily basis.

H. Top stations

Question: *“Which stations are frequented by different users?”*

```
#Casual's top stations
casual_stations = (df[df["member_casual"] ==
"casual"]["start_station_name"].value_counts().head(10))
```

	count
start_station_name	
Unknown_Station	378958
DuSable Lake Shore Dr & Monroe St	30796
Navy Pier	26971
Streeter Dr & Grand Ave	23367
Michigan Ave & Oak St	22123
DuSable Lake Shore Dr & North Blvd	19071
Millennium Park	18819
Shedd Aquarium	16553
Theater on the Lake	15491
Dusable Harbor	15446

➩ **Casual riders users primarily frequent stations located near tourist and leisure areas, suggesting recreational use of the service.**

```
#Member's top stations
member_stations = (df[df["member_casual"] ==
"member"]["start_station_name"].value_counts().head(10))
```

	count
start_station_name	
Unknown_Station	716363

Kingsbury St & Kinzie St	30894
Clinton St & Washington Blvd	25304
Canal St & Madison St	21843
Clinton St & Madison St	21820
Clark St & Elm St	20607
State St & Chicago Ave	18827
Wells St & Elm St	18667
Wells St & Concord Ln	18110
Clinton St & Jackson Blvd	18039

⇨ **Members primarily use stations located near offices, train stations, and financial centers, suggesting business use.**

An interactive dashboard has been created on Tableau Public :
https://public.tableau.com/shared/CPSKRYTXK?:display_count=n&:origin=viz_share_link

KEY INSIGHTS

The analysis clearly shows that members use the bike-sharing service more frequently than casual users. Their usage is significantly higher during weekdays, with noticeable peaks around 8 a.m. and 5 p.m., suggesting that members primarily use the service for commuting and daily transportation purposes. This interpretation is further supported by the observation that many frequently used stations are located near train stations, business districts, and office areas. Usage decreases during weekends, reinforcing the idea of routine-based travel behavior.

Casual users, on the other hand, represent a smaller share of total rides but generally take longer trips and show higher activity during weekends. This pattern suggests a more recreational use of the service. This interpretation is also supported by the fact that many frequently used stations appear to be located near tourist attractions, leisure areas, and activity centers.

The analysis also revealed a seasonal pattern in service usage, with a noticeable increase among casual users between June and August. This period corresponds to warmer months and summer holidays, suggesting that weather conditions and seasonal activities may strongly influence bike usage.

Finally, the analysis indicates a higher overall usage of electric bikes. This may suggest that users perceive electric bikes as a more convenient and comfortable option, requiring less physical effort while providing greater flexibility for both short and long-distance trips.

RECOMMENDATIONS

1. ***Develop leisure-oriented membership plans***

Rationale:
The analysis showed that casual riders tend to take longer trips and display higher activity during weekends, suggesting a predominantly recreational use of the service.

Proposed actions:

- Introduce **Weekend Membership Plans** designed for occasional leisure riders.
- Offer **Leisure Passes** targeted at users who mainly ride during weekends or holidays.
- Create **Family Packages** allowing multiple users to share benefits at discounted rates.

Expected impact:
These offers could encourage casual users to transition from one-time usage to recurring subscriptions.

2. ***Strengthen commuting-focused marketing for members***

Rationale:
Members show clear weekday peaks around morning and evening hours, indicating routine commuting behavior.

Proposed actions:

Marketing campaigns emphasizing convenience and savings: *"Save money on your daily commute!"*

Additional messaging could focus on:

- Cost savings compared with alternative transportation
- Time efficiency
- Flexibility for daily travel

Expected impact:
Increase retention and reinforce the value of membership among existing users.

3. ***Launch seasonal campaigns***

Rationale:
Bike usage increases significantly during summer months and decreases during winter, suggesting strong seasonality effects.

Proposed actions:

Summer campaigns

- Introduce a **Summer Membership**
- Offer a **Summer Pass**
- Create limited-time promotions during peak seasonal demand

Winter campaigns

- Introduce rewards and discounts
- Create winter riding challenges
- Offer seasonal incentives to maintain engagement

Expected impact:
Capture high-demand periods while reducing seasonal declines in usage.

4. ***Promote e-bike incentives for membership conversion***

Rationale:
The analysis indicates a stronger preference for electric bikes among casual users.

Proposed actions:

- Offer e-bike credits for new members
- Provide discounts on electric bike usage after subscription
- Include free e-bike trials in membership packages

Expected impact:
Encourage casual riders to subscribe by leveraging existing preferences.

5. ***Implement a "Ride & Earn" loyalty program***

Rationale:
Casual riders mainly use the service for leisure activities during weekends, while members show more routine commuting patterns. A rewards system could increase engagement among existing members and encourage casual users to subscribe.

Proposed actions:

Ride & Earn system

- 1 ride = X points (+ weekend bonus)
- 10 rides = free riding minutes
- 25 rides = e-bike upgrade
- Weekend bonus challenges
- Casual riders may accumulate points, but points can only be redeemed after becoming members

Expected impact:

- Increase weekend engagement among members
- Improve customer retention
- Encourage casual-to-member conversion
- Increase overall usage frequency

CONCLUSION

This analysis highlighted clear behavioral differences between members and casual riders. Members use the bike-sharing service more frequently and mainly during weekdays, with usage peaks during commuting hours, suggesting routine and transportation-oriented behavior. In contrast, casual riders tend to take longer rides, show stronger weekend activity, and display more pronounced seasonal patterns, indicating a more recreational use of the service.

These findings suggest that converting casual riders into members requires targeted strategies aligned with their usage habits and preferences. Leisure-oriented subscriptions, seasonal offers, e-bike incentives, and loyalty programs such as a reward system could encourage more regular engagement and increase membership adoption.

By leveraging data-driven insights, the company can develop personalized marketing strategies aimed at improving customer retention, increasing service usage, and supporting long-term growth.

Future analyses could incorporate weather data and customer demographics to gain a deeper understanding of user behavior.